

Project Progress and Experimental Design

Project Aim

This project investigates length generalisation failure in Transformer-based sequence models. The main question is:

When a Transformer is trained on short sequences, what fails when it is tested on much longer sequences, and can these failures be systematically classified?

The project focuses on controlled synthetic tasks rather than natural language data so that sequence length, target outputs, and failure points can be measured precisely.

Experimental Design

The experiments follow a train-short/test-long setup.

Models are trained on short or moderate sequence lengths and evaluated on longer unseen lengths up to 1024. This allows us to test whether the model has learned a length-general algorithm or only a length-specific pattern.

The main tasks used are:

- **Addition**: tests arithmetic reasoning and carry propagation.
- **Delayed copy**: tests long-range memory after a separator token.
- **Reverse**: tests precise positional remapping.
- **Associative recall**: explored during validation, but not included in the final full matrix because the main robust results focus on addition, delayed copy, and reverse.

The original simple copy task was replaced with delayed copy because simple copy was too easy and did not meaningfully test long-range memory.

Positional Encodings Compared

The project compares several positional encoding strategies under the same training and evaluation setup:

- Learned absolute positional embeddings
- Sinusoidal positional encoding
- No positional encoding
- ALiBi
- RoPE

This comparison was included because the literature suggests that positional representation strongly affects length extrapolation.

Experiments Completed

The main experimental matrix is complete:

- 3 main tasks: addition, delayed copy, reverse
- 5 positional encoding methods
- 3 random seeds: 42, 123, 2024

This gives 45 main experimental runs.

For each run, results are reported across increasing test lengths. The results include:

- token accuracy
- exact match accuracy
- mean across seeds
- standard deviation
- standard error
- failure length
- generalisation gap

Main Findings So Far

The central result is that models often fit the training length but fail when evaluated on longer lengths.

For example:

- Addition trained at length 16 reaches high exact match at length 16, but drops to 0% at length 32 and remains 0% up to 1024.
- Delayed copy trained at length 128 reaches near-perfect exact match at length 128, but drops to 0% at length 256 and beyond.
- Reverse trained at length 128 also reaches 100% exact match at length 128, but drops to 0% at length 256 and beyond.

This suggests that the models are not learning robust length-general algorithms.

ALiBi and NoPE Observation

ALiBi and NoPE were checked carefully because some earlier results looked suspicious.

After correcting the positional encoding implementation, ALiBi and NoPE often fail even at the training length. Therefore, in this setup, they should be discussed mainly as training-length underfitting cases, not as successful models that fail only during extrapolation.

This distinction is important for the failure taxonomy.

Attention-Based Failure Analysis

To understand why models fail, attention diagnostics were added.

The attention analysis compares model behaviour at:

- the training length
- the first failure length
- length 1024

The diagnostics include:

- attention entropy
- normalised attention entropy
- average attention distance
- local attention ratio

The main pattern is that attention becomes more diffuse at longer lengths. Normalised attention entropy increases and local attention ratio decreases, while exact-match accuracy collapses.

This suggests that attention patterns learned at short lengths do not transfer reliably to longer sequences.

Mitigation Experiments

Several simple mitigation strategies were tested:

- mixed-length training
- curriculum training
- randomised padded mixed-length training

The results show:

- Mixed-length training can cause underfitting.
- Curriculum training restores or preserves train-length accuracy.
- Randomised padded training also preserves train-length accuracy.
- However, none of these strategies improve exact-match extrapolation at 256, 512, or 1024.

This suggests that simply exposing the model to multiple shorter lengths is not enough to make it learn a length-general algorithm.

Failure Taxonomy

Based on the results, five failure types have been identified:

1. Extrapolation collapse

The model fits the training length but exact-match accuracy collapses outside the training range.

2. Training-length underfitting

The model fails even at the training length, so it cannot be treated as a meaningful extrapolation test.

3. Seed instability

Some task and positional encoding combinations vary strongly across random seeds.

4. Attention diffusion

Attention becomes less structured as sequence length increases.

5. Mitigation-resistant failure

Simple mitigation strategies preserve train-length performance but do not improve long-length extrapolation.

Current Interpretation

The results currently support the view that the tested Transformer models learn length-specific solutions rather than robust algorithms.

The failure is not only caused by one positional encoding, one random seed, or one task. It appears across tasks, positional encodings, seeds, attention diagnostics, and mitigation attempts.

Feedback

I would like feedback on the following points:

1. Is it acceptable that associative recall was explored but not included in the final full matrix?
2. Should ALiBi and NoPE be discussed as underfitting cases?
3. Are the five failure categories clear and well separated?
4. Are the mitigation experiments sufficient, or should one small additional mitigation experiment be added?

Current Project Status

I have made a rough draft for the report on these findings and I will update it with your feedback and share it.